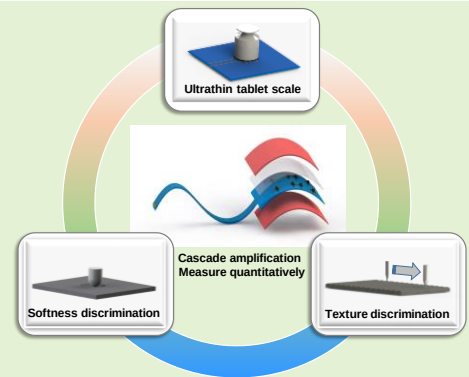


Cascade Amplification Effect for Mechanical Stimuli Sensors by Designing the Current Path Through Carbon Fiber Beams

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Abstract—Inspired by the tandem structure can be used as an effective approach to improving the performance of optoelectronic devices such as solar cells and organic light emitting diodes, a series of mechanical sensors with high performance were fabricated on a flexible sheet (thickness, less than 300 μm) based on tandem structures. In our designed tandem structure models, the single sensing unit consisting of one single carbon fiber beam presents nonlinear behavior when responding to force stimuli. Thus, the initial resistance and resistance change value of the tandem mechanical sensors could be enhanced by increasing the number of sensing units to lengthen the current path through carbon fiber beams and enable the sensor to respond to subtle force after preloading a large force. Moreover, the tandem mechanical sensors can measure force quantitatively due to the corresponding relationship between the resistance change rate and the loading force. These combined advantages of the tandem mechanical sensors were also demonstrated by the successful application in weighed, softness discrimination and texture discrimination, implying that these mechanical sensors could be used as mechanoreceptors like a finger touching in monitoring daily life sensations.

Index Terms—Mechanical sensors, Tandem structure, Ultrathin tablet scale, Softness discrimination, Texture discrimination



I. INTRODUCTION

The fingers, as the extremities of the human hand, especially the fingertips are covered with denser nerve endings than skin in many other parts of the human body and can be sensory

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receptors of pain, cold, warm, and mechanical stimuli, making the human hand the most perfect tool that nature can create [1]–[3]. However, the fingers cannot visualize and digitize the sensory information that responds to stimuli from the outside environment. Therefore, bionic fingers have been developed to visualize sensory information [4]–[6]. An essential visualization process involves accurately transducing physical signals into electronic signals, which entails applying various kinds of sensors [7]–[12]. Among these sensors, mechanical sensors that can respond to mechanical stimuli such as static forces, stretch strain, vibrations, and object shape and texture play a crucial role in visualizing sensory information [13], [14].

Various flexible mechanical sensors have recently been widely investigated and have made significant progress [15]–[18]. The development route of flexible mechanical sensors can be categorized into two ways: active materials and structure designs [19]–[21]. The most commonly used active materials are carbon-based materials (e.g., graphene, CNTs, and carbon fibers), conductive polymers (e.g., PEDOT: PSS, PVDF, PPy, and hydrogels), and metal-based materials (e.g., liquid metals, metallic nanoparticles and nanowires) [22]–[32]. These active materials have exhibited excellent electrical conductivity, good mechanical properties, and long-term stability, enabling the mechanical sensors based on these active materials to possess excellent performance with high sensitivity, a more extensive working range, and reliable stability. Among these active materials, carbon fibers have the extra advantages of being inexpensive, high available, and free-standing, making

them robust candidates to fabricate high performance mechanical sensors [24]. In addition to developing active materials, structure designs have played a remarkable role in flexible mechanical sensors. Mechanical sensors with structures such as meshes, waves, ribbons, 3D spirals, serpents, microspheres, pyramids, and fabrics have been developed, which enable mechanical sensors with high-sensitivity and excellent durability [33–40]. However, most of these structures are low-yield and involve a complicated multistep fabrication process. It is still a challenge to develop a simple structure design approach for high-performance mechanical sensors.

Researchers were not content with mechanical sensors that only responded to a single mechanical stimulus, they were also devoted to developing mechanical sensors with simultaneous multi-stimuli detection capability. For example, Pang et al. fabricated a mechanical sensor based on a graphene/polymer composite with a foam-structure to achieve pressure and strain sensing properties [17]. Yeo et al. utilized a microfluidic manifold with a 3D elastomeric architecture to create a wearable mechanotransduced tactile sensor to detect both normal and shear forces and the discrimination of surface features, contour changes, and object stiffness [18]. These previous studies have demonstrated that mechanical sensors could be promising candidates to mimic the mechanoreceptors of human fingers to visualize sensory information. However, there are three performances aspects requiring improvement for the mechanical sensors: 1) The sensitivity in the relatively large force range needs to be further improved for appropriate responses to the mechanical stimuli; 2) The sensory information can be described quantitatively, not qualitatively; 3) Device structure should be able to tune as required in actual applications.

Aiming at improving these performances, we developed a novel mechanical sensing unit assembled by only one carbon fiber beam with a diameter of 300 μm and a length of 3 mm. We cascaded two or more sensing units to form a series of tandem structures to increase the times of the current path through the sensing layer (carbon fiber beams) to magnify the change in resistance and enhance the sensitivity in the relatively large force range. The corresponding relationship between the resistance change rate and the loading force was constructed to offer the quantitative measurement of the mechanical sensors. Moreover, the tandem structure of mechanical sensors can be tuned by regulating the number of sensing units to make them suitable for different applications such as weighing, softness discrimination and texture discrimination.

II. RESULTS AND DISCUSSION

A. Tandem structure Model for Mechanical Sensors

Fiber-based materials are often designed and synthesized with characteristics such as high strength, high stretchability, or good chemical stability, which have been widely investigated [41]–[43], while the characteristics of free-standing materials are often ignored. Focusing on free-standing fiber-based materials, we design a series of tandem structure models for mechanical sensors. As shown in Fig. 1(a), a single sensing unit is composed of one single conductive carbon fiber beam crossing over the bottom Cu electrode layer, in which the carbon fiber beam has a diameter of about 300 μm (Fig. S1) and a length of 3 mm. The electronics can transport from the Cu

electrode to the carbon fiber beam under an applied voltage and form a loop. According to our published reference [44], the carbon fiber beam comprises thousands of carbon fibers, and there are thousands of gaps between the carbon fibers. When pressure is applied to the carbon fiber beam, the size and the total number of gaps decrease, allowing the electronics to be transported through the carbon fiber beam more easily, resulting in a decrease in the transverse resistance of the carbon fiber beam. Therefore, the part of the carbon fiber beam contacting the Cu electrode could be equivalent to a variable resistance in the single sensing unit. The resistance of the single sensing unit response to force is nonlinear, and tends to saturate under large forces (Fig S2a), indicating that the poor sensitivity of the single sensing unit when a large force is applied. To sensitivity respond to a subtle force ever after a large force had been loaded, a series of tandem structure models were designed by increasing the number of sensing units, such as cascades of two sensing units, four sensing units, and eight sensing units. The initial resistance of these tandem structure models increases exponentially with the number of sensing units due to the current path through which carbon fiber beams are increased. In addition, the longitudinal section of carbon fiber beams compresses at different levels when a constant force is applied on the surface of these tandem structure model based mechanical sensors. Similar to a proverb, "a chopstick is easy to break, while a bunch of shopsticks are not," one single sensing unit that consists of a single carbon fiber beam is easily compressed to saturation, and greater force needs to be applied as the number of sensing units increases. Therefore, the mechanical sensors with more sensing units could possess a larger initial resistance (R_0), greater resistance change value(ΔR), and can respond to a subtle force after a large force has been loaded.

Fig. 1(b) shows the architectonics of the mechanical sensors, which were constructed of a sensing layer, bottom encapsulation layer, and top encapsulation layer. The sensing layer is based on the tandem structures above, eight carbon fiber beams are crossed over their corresponding Cu electrode layers and fixed on the FPCB through copper holes taking the tandem structure with eight sensing units as an example. The bottom encapsulation layer was composed of one single layer of PET, while two layers of PI and PET formed the top encapsulation layer. Both the bottom encapsulation layer and the top encapsulation layer were used to protect the sensing layer. When applying continuously increasing force on these cascade mechanical sensors, the response in resistance variation is shown in Fig 1(c). The resistance change value of the mechanical sensor with one sensing unit reached saturation under a force of 471 mN, and the saturated value of ΔR was 47.5 Ω . As the number of sensing units increased, the sensors reached saturation at a much larger force and obtained a more considerable saturated value of ΔR . Notably, the resistance change value of the sensor with eight sensing units continues to increase under forces from 0 mN to 2500 mN, and the value of ΔR is much larger than the other sensors with fewer sensing units when a little bit more force is applied. Therefore, in contrast to the sensor with one sensing unit, the sensor with eight sensing units can still respond to subtle force after preloading a large force. As shown in Fig 1(d), the sensor with eight sensing units presented a significant change in resistance

after preloading a force of 446 mN, while the sensor with one sensing unit could not observe a noticeable change in resistance. In particular, the sensor with eight sensing units can respond to another 20 mN ever after it has been loaded 645 mN,

demonstrating the cascade amplification effect of the sensor with tandem structure. Meanwhile, mechanical sensors with a significant change in resistance have great potential in practical applications.

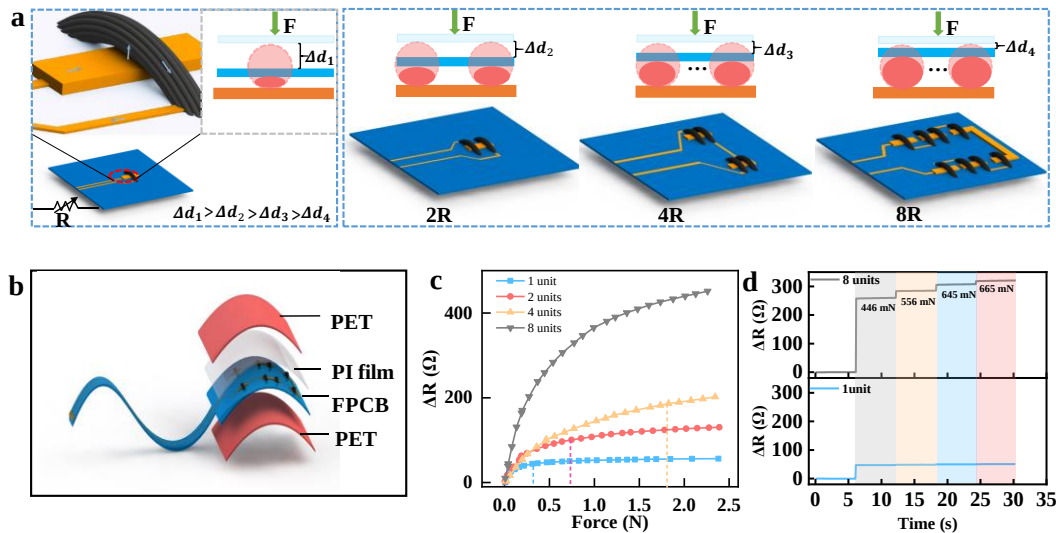


Fig. 1. (a) Tandem structure models with single sensing unit, two sensing units, four sensing units, and eight sensing units, respectively. (b) Architectonics of the mechanical sensor with tandem structure. (c) Responses in resistance of the mechanical sensors with single sensing unit, two sensing units, four sensing units, and eight sensing units when applied continuously increasing force, respectively. (d) Resistance change value of the mechanical sensors with single sensing unit and eight sensing units when respond to a series of extra increasing force after loaded a force of 446 mN, respectively.

B. Performance Characterization of Mechanical Sensor

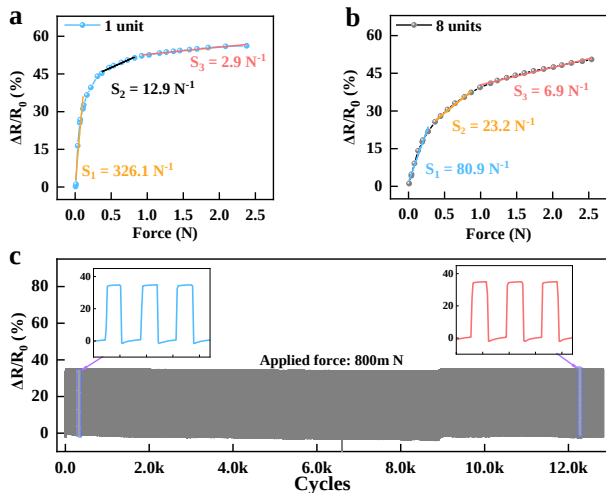


Fig. 2. (a) Resistance change rate of the mechanical sensor with single sensing unit response to the loading force. (b) Resistance change rate of the mechanical sensor with eight sensing units response to the loading force. (c) Responses in resistance of the mechanical sensor with eight sensing units under repeated loading-unloading over 10000 cycles.

Sensitivity reflects the capacity for appropriate responses to mechanical stimuli is a basic parameter for the performance characterization of mechanical sensors [45]–[47]. Mechanical stimuli such as texture, size, shape, hardness and softness of the measured object could be obtained through other mechanical stimuli by static force with high sensitivity and high resolution [48]–[50]. Therefore, the sensitivity and resolution of the

mechanical sensor could be demonstrated by responses to the static force. In this contribution, the sensitivity of the pressure sensors is defined as

$$S = \frac{d(\frac{\Delta R}{R_0})}{dF}, \quad (1)$$

where the F is the loading force. Fig 2(a) presents the resistance change rate of single sensing unit responses to the force, and there is a nonlinear relationship between the resistance change rate and the loading force. A high sensitivity of 326.1 N^{-1} was obtained for the single sensing unit when the loading force was in the range of 0–0.12 N. The sensitivity dropped drastically when the force was over 0.12 N, a sensitivity of 12.9 N^{-1} in the force range of 0.35–0.85 N, and a low sensitivity of 2.9 N^{-1} corresponded to the force range of 0.9–2.4 N. In contrast to the single sensing unit, the tandem structured mechanical sensor with eight sensing units shows a slightly lower sensitivity of 80.5 N^{-1} in the force range of 0–0.28 N (Fig. 2(b)), but this sensor showed a sensitivity of 23 N^{-1} in the force range of 0.35–0.85 N, higher than the sensor with single sensing unit, demonstrating that the tandem structure can improve the sensitivity of the mechanical sensor in the relatively large force range. In practical applications for object softness and texture detection, the mechanical sensor is loaded with a large force once it touches the surface of the object, and then a small force occurs at the interface when the sensor acts on the object by the formalization of point-touch or linear motion for the measurement. Therefore, the sensor with eight sensing units can recognize a subtle force after preloading a large force is ideal for the feature characterization of the measured object. Durability also plays a vital role in the practical application of mechanical sensors [51], thus, the durability of the mechanical

sensor was performed under repeated loading-unloading cycles of an external force of 800 mN (as shown in Fig. 2(c)). The resistance change rate of the sensor remained negligible change and could still return to the initial value after 10000 cycles, indicating the excellent durability of the mechanical sensor during loading-unloading cycles. A video was also collected to vividly assess the real-time evolution of loading-unloading cycles of the mechanical sensor (Movie S1).

C. Ultrathin Table Scale and Quantitative Force Measurement

The mechanical sensors were made up of one thin sensing layer and two thin encapsulation layers as described above, and their total thickness was approximately 256.5 μm (as shown in Fig. S3) (the carbon fiber beams were tensed and compressed), which was obtained by the 3D shooting of Leica 3D optical microscopes (Leica DVM6). Moreover, the tandem structure with eight sensing units distributed on the substrate in a U-shape makes the mechanical sensors relatively flat, the mechanical sensor with eight sensing units is more suitable for use as an ultrathin tablet scale (different batch from part A) (Fig. 3(a)), and can give the ultrathin tablet scale the inherent advantages of portability and low cost. In addition, the weight of the measured object could be inferred from the resistance change rate of the mechanical sensors. As shown in Fig. 3(b), the relationship between the resistance change rate and the loading force is consistent with the fitted curve based on an exponential function

$$F = A * (\exp(B * \frac{\Delta R}{R_0}) - 1), \quad (2)$$

the loading force is corresponds to the change resistance rate. Meanwhile, this relationship can also be applied to the mechanical sensor with a single sensing unit (Fig. S4). Thus, the loading force amount could be obtained quantitatively through the mechanical sensors, indicating the great potential of application in weight measurement. The mechanical sensors measured seven weights with different marked weight of 2, 5, 10, 20, 30, 50, and 100 g. Fig. 3(c) shows that the measured weight of these weights were 2.2, 5.2, 10.3, 21.7, 30.8, 51.8 and 104.5 g, respectively, which were obtained according to the above relationship. Moreover, there was only a slight deviation between the measured weight and the marked weight (Fig. 3(d)), demonstrating that the mechanical sensors with eight sensing units could be used as a portable ultrathin tablet scale to apply weight measurements (see Movie S2). In addition to the mechanical sensor with eight sensing units, the sensor with single sensing unit, two sensing units, and four sensing units were also try to use as ultrathin tablet scale. These mechanical sensors measured the weights with marked weight of 10, 20, 50 g, respectively. Unfortunately, the measured weight from the sensor with single sensing unit and two sensing units showed significant error to the marked weight (Fig. S5(a), S5(b), S5(c), S5(d)), which may be attributed to the bottom area of the weights are much larger than the active area of these sensors and the irregular interface between the device and the weights, resulting in the weights tipped to one side. The measured weight from the sensor with four sensing units presented less error than the data from the sensors with single sensing unit and two sensing units (Fig. S5(e), S5(f)), but not nearly as accurate

as the sensor with eight sensing units (Fig. 3(c), 3(d)), further demonstrated the mechanical sensor with eight sensing units is more suitable for use as an ultrathin tablet scale.

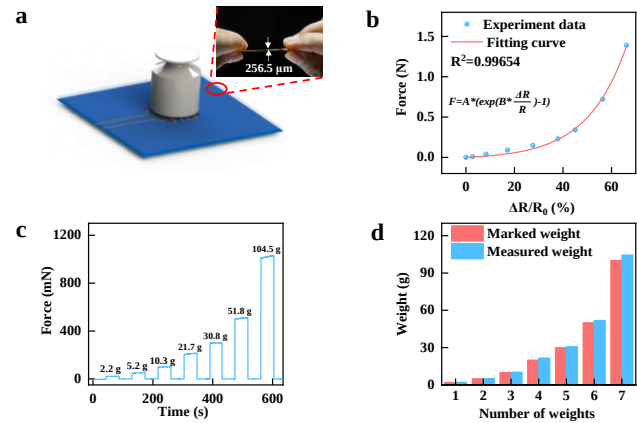


Fig. 3. (a) Schematic diagram of ultrathin tablet scale. (b) Relationship between the resistance change rate and the loading force. (c) Measured weight of the weights obtained by the mechanical sensor with eight sensing units. (d) Comparison of the measured weight and marked weight of the weights.

D. Softness Discrimination

The mechanical sensors with the combined advantages of high sensitivity, high resolution, and the ability to measure force quantitatively can characterize the feature of measured object. The shape, texture and hardness of the measured object have been widely investigated through various sensors [52]–[55], while the softness has rarely been reported among the feature characterization of the measured object. In contrast to the hardness, which the measured object is usually biased toward rigid materials, such as metal [56], the softness represents the level of the soft, smooth, and flexure of the measured object, and is mainly used to characterize the fabric and silica gel in the industry. For example, a two-component epoxy adhesive (AB glue), which was cured by mixing epoxy resin and curing agent at different ratios, shows various softnesses but identical appearances after curing. Therefore, the mechanical sensors similar to the mechanoreceptors of human skin may be an ideal option for the softness discrimination of AB glue. As shown in Fig. 4(a), the mechanical sensors were used to detect the softness of AB glues, in which the AB glues were mixed with different ratios of 20:0.8, 20:1, 20:1.2, and 20:1.5, respectively. It could be observed that there were no significant differences in the appearances of these AB glues (see Fig. S6). According to the different measuring mechanisms, there are two approaches for the softness discrimination of AB glue through the mechanical sensors. The first approach relies on the corresponding relationship between the resistance change rate and the loading force. A 3D mobile control optical platform was used to control the mechanical sensor to press the surface of the AB glues (Fig. S7), and stop pressing down when the resistance change rate achieved a set value. This means that an equal force was applied in the first approach, named the constant force mode. Fig. S8 shows the resistance change rate of the mechanical sensor under press down on the AB glues and lift with four cycles. Although the waveforms of these resistance change rates show similar

morphologies, the resistance change rate exhibited a significant difference in the slope during the pressing process (as shown in Fig. 4(b)). The slopes increased with the curing agent content, for example, the AB glues ratios were 20:0.8, 20:1, 20:1.2, and

20:1.5, and the corresponding slopes of about 0.0259, 0.0264, 0.0285, and 0.0295, respectively (Fig. 4(c)). Therefore, the softness of the measured object could be distinguished by the slopes of the resistance signal under constant pressure.

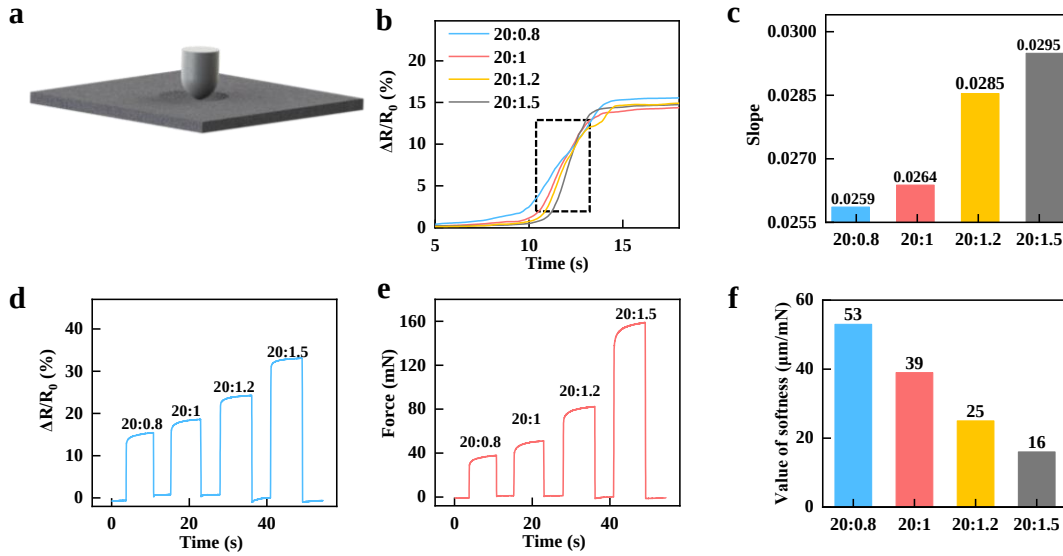


Fig. 4. (a) Schematic diagram of softness discrimination by the mechanical sensor. (b) Responses in resistance of the mechanical sensor when press the surface of the AB glues with a constant force of 35 mN. (c) Slope of the resistance change rate for the mechanical sensor when press on the AB glues. (d) Responses in resistance of the mechanical sensor when press the surface of the AB glues with a constant height. (e) Responses in force of the mechanical sensor when press the surface of the AB glues with a constant height. (f) Value of softness for the AB glues with different ratios.

In contrast to the first approach, the second approach is based on the constant height mode, in which the mechanical sensors press down on the AB glues through a constant height. As shown in Fig. S9, the mechanical sensor was used to recognize the softness of the measured object with the assistance of a robot (B005). This robot can control the sensor trajectory, and ensure that the sensor presses down to the same height each time. Fig. S10 presents the real-time evolution in the resistance signal of the mechanical sensor when pressed down on the AB glues. It can be observed that the resistance drops more drastically as the ratios of the AB glues increase, indicating less softness as the content of the curing agent increases. The harder surface increases the compression of the sensing layer of the mechanical sensor, and then the gaps between the carbon fibers decreased, resulting in a decrease in the resistance of the sensor. In contrast to the first approach, the second approach can distinguish softness more intuitively. As shown in Fig. 4(d), the softness of the AB glues could be inferred directly from comparing the resistance change rate. According to the corresponding relationship between the resistance change rate and the loading force, more force was applied as the softness of AB glues decreased (Fig. 4(e)). Therefore, the mechanical sensors with less sensing unit may cannot distinguish the different softness because they respond to a force of over 0.12 N with a relative low sensitivity. As shown in Fig. S11, the resistance change rate of the sensor with single sensing unit that correspond to the AB glues ratios of 20:1.2 and 20:1.5 are very close, this also happened in the sensor with two sensing units.

Although the sensor with four sensing units shows a trend of increase of the resistance change rate when the AB glues ratios increase from 20:08 to 20:1.5, but not as significant as the sensor with eight sensing units, indicated that the mechanical sensor with eight sensing units may be the most suitable candidate for softness discrimination. Moreover, there is a compressive strain at the contact interface between the AB glues and the mechanical sensor during press down, so the value of softness can be defined as the strain distance corresponding to the unit force. The relationship between the value of softness and the force can be expressed as

$$S = \frac{\Delta d}{F}, \quad (3)$$

where S is the value of softness, Δd is the distance of compressive strain, which is equal to $t v$, and here t is the time from the sensor contacting the surface of the AB glues to leave the surface, and v is the velocity of the mechanical sensor when pressing down on the surface of the AB glues. Therefore, the value of softness of these AB glues can be obtained by the (3), as shown in Fig 4(f), the softness values are 53, 39, 25, and 16 $\mu\text{m/mN}$ for the corresponding AB glues, respectively, consistent with the lower softness as the content of the curing agent increases. A video was also collected to vividly demonstrate the second approach for softness discrimination (Movie S3). Together, these two approaches offer simple and feasible alternate schemes for the softness detection of fabric and silica gel.

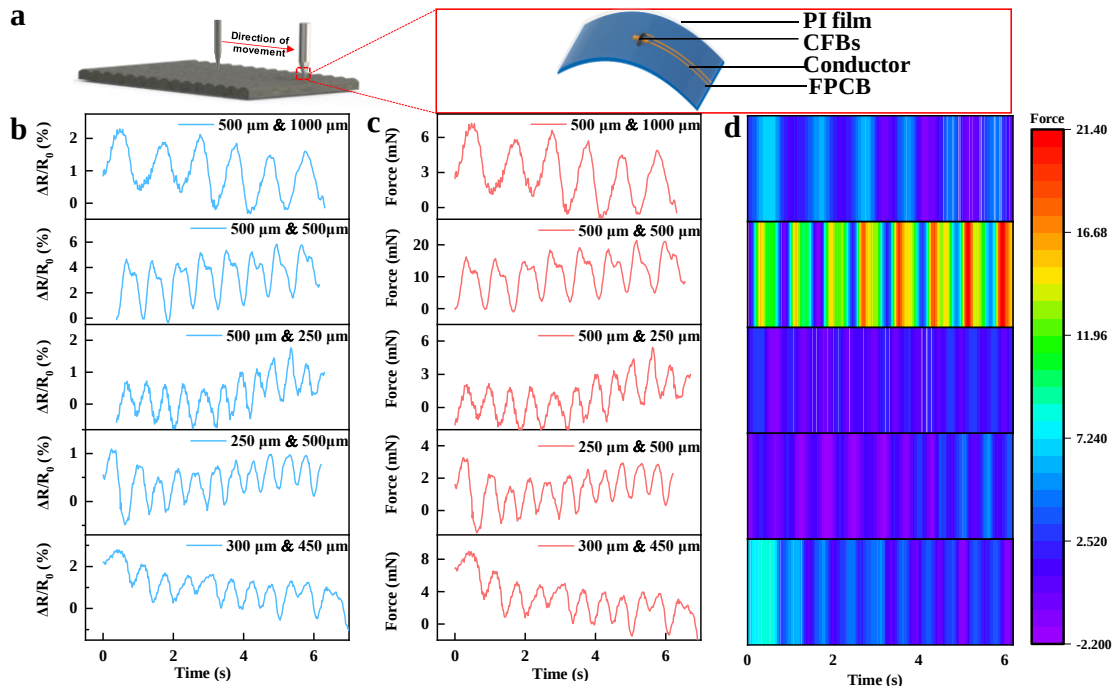


Fig. 5. (a) Schematic diagram of texture discrimination by the mechanical sensor with single sensing unit. (b) Responses in resistance of the mechanical sensor when scan on the surfaces with different periodical fringes. (c) Responses in force of the mechanical sensor when scan on the surfaces with different periodical fringes. (d) Distribution of the loading force corresponding to the resistance of the mechanical sensor when scan on the surfaces with different periodical fringes.

E. Texture Discrimination

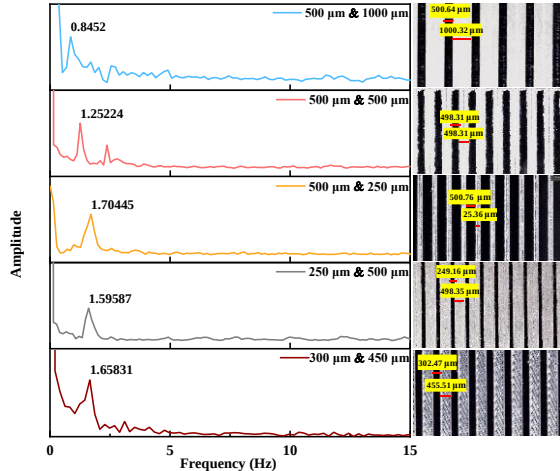


Fig. 6. Fast Fourier transform (FFT) corresponding to the resistance of the mechanical sensor when scan on the surfaces with different periodical fringes.

In addition to the softness, the texture of the measured object was also performed by the mechanical sensors. The sensor was driven by a single axis step motor with the same scanning speed of 1.25 mm/s for the objective characterization of the texture of the measured objects (Fig. S12), in which the surfaces of the measured objects were prepared with different periodical fringes by a 3D printer. As shown in Fig. 5(a), the periodical fringes were assumed with different line widths and groove widths of 500 & 1000, 500 & 500, 500 & 250, 250 & 500, and 300 & 450 μm . Unlike the applications of ultrathin table scale

and softness discrimination, the texture discrimination requires the mechanical sensor with a small size probe tip in addition to high-sensitivity. As shown in Fig S13(a), the mechanical sensors with single sensing unit, two sensing units, and eight sensing units all can recognize the detailed information of the large periodical fringes with line width and groove width of 500 & 1000 μm . When the line width and groove width of the periodical fringes decrease to 500 & 250 μm , the mechanical sensor with single sensing unit still can recognize the detailed information, while the sensors with two sensing units and eight sensing units have encountered missing information (Fig S13(b)). These results illustrated that the mechanical sensor with single sensing unit is suitable for texture discrimination because the small tip size probe tip of sensor can obtain higher resolution. The resistance change rate of the mechanical sensor response to the different periodical fringes showed the sensor with the capacity of texture discrimination (Fig. 5(b)), which was also demonstrated with a video (Movie S4). Moreover, the distribution of the loading force corresponding to the periodical fringes was also obtained (Fig. 5(c), 5(d)), which could describe the cyclical variations in the surface of the measured object reliably and intuitively. In addition, the resistance change rate curve could be transformed into fast Fourier transform (FFT) to demonstrate the capability of texture discrimination of the mechanical sensors (as shown in Fig. 6). The frequency peak positions redshifted as the groove width decreased and showed negligible variation when the total value of the line width and groove width remained unchanged, indicating that the mechanical sensors can be used to discern the surface texture of the measured object. Furthermore, the surface with the periodic fringes of line width and groove width of 500 & 500 μm was also scanned by the other three different

velocities of 0.25, 0.5, and 2.5 mm/s. As shown in Fig. S14, the frequency peak value increases linearly with the scanning velocity, consistent with that previously reported by Cao et al. [57]. The mechanical sensor can recognize the periodical fringes and discriminate the detailed texture by the distribution of the loading force. Overall, the successful application of the sensor with single sensing unit in texture discrimination further demonstrated that the mechanical sensors can be tuned by regulating the number of sensing units to make them suitable for different applications.

III. CONCLUSION

In summary, we developed a series of mechanical sensors with tandem structures by cascading two or more sensing units. Both the initial resistance and resistance change values improved as the number of sensing units increased. In contrast

to recently reported mechanical sensors, the sensor with a tandem structure can maintain a moderate sensitivity in the force range of 0.35-0.85 mN (Table S1), enabling the sensor to respond to subtle forces after preloading a large force. The sensors can also measure the force quantitatively by establishing the corresponding relationship between the resistance change rate and the loading force. The mechanical sensors attribute to these excellent characteristics could be used as an ultrathin tablet scale for weighing, used as bionic fingers for softness discrimination, and texture discrimination. These successful applications also demonstrated that the mechanical sensors could make bionic fingers be perfect tools that are similar to real human fingers.

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